

Chapter 22 - Dynamics

1. Dynamics
 - a. Kinematics
 - b. Kinetics
 - c. Basic Terms
 - i. Motion
 - ii. Rest
2. Motion of a particle in a Straight Line
 - a. Uniform Motion
 - i. Average Velocity
 - ii. Instantaneous Velocity
 - b. Uniformly Accelerated Motion
 - i. Average Acceleration
 - ii. Instantaneous Acceleration
3. Motion with Uniform Acceleration
 - a. First equation of motion
 - b. Second equation of motion
 - c. Third equation of motion
 - d. Distance travelled at t^{th} second
4. Motion under gravity
5. Motion Down a Smooth Inclined Plane
 - a. Equations for body sliding downwards
 - b. Equations for body climbing (or moving) upwards

#Dynamics

Dynamics is the branch of mechanics that deals with the study of bodies at motion. Dynamics can be further divided into two parts: Kinematics and Kinetics.

Kinematics	Kinetics
Kinematics is the branch of dynamics that deals with the study of bodies in motion with no consideration of forces producing motion.	Kinetics is the branch of dynamics that deals with the study of bodies in motion together with the forces causing the change in the state of motion.

Basic Terms

Motion	A body is said to be in motion if it changes its position with respect to the time. If the path of the moving body is a straight line then the motion is said to be a rectilinear motion.
Rest	A body is said to be at rest if it does not change its position with respect to the time.

#Motion of a particle in a Straight Line

Uniform Motion

Uniform motion is defined as the motion of an object traveling in a straight line at a constant speed, covering equal distances in equal time intervals. It is characterized by zero acceleration, as the velocity remains unchanged.

Average Velocity	Instantaneous Velocity
The ratio of the total change in displacement to the total time taken, is called average velocity.	The velocity of a body at a particular instant of time or position, is called instantaneous velocity.
It is denoted by $v_{av} = \frac{\Delta x}{\Delta t}$	It is denoted by $v = \frac{dx}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}$

Uniformly Accelerated Motion

Uniformly accelerated motion is a type of motion where an object's velocity changes by equal amounts in equal intervals of time. This means the object is speeding up or slowing down at a constant rate, resulting in a constant acceleration.

Average Acceleration	Instantaneous Acceleration
The ratio of the total change in velocity to the total time taken, is called average acceleration.	The acceleration of a body at a particular instant of time or position, is called instantaneous acceleration.
It is denoted by $a_{av} = \frac{\Delta v}{\Delta t}$	It is denoted by $a = \frac{dv}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}$

Note: We can also express the acceleration as $a = \frac{d^2x}{dt^2}$ and $a = v \frac{dv}{dx}$

#Motion with Uniform Acceleration

First equation of motion

Let u and v be the velocities of a body at time 0 and t respectively. Thus, by definition, acceleration is

$$a = \frac{dv}{dt}$$

or, $dv = a dt$

Integrating within the limits provided, we have

$$\int_u^v dv = \int_0^t a dt$$

or, $[v]_u^v = a [t]_0^t$

or, $v - u = at$

$\therefore v = u + at$

Second equation of motion

Since we have $v = u + at$

using $v = \frac{dx}{dt}$, we rewrite it as

$$\frac{dx}{dt} = u + at$$

or, $dx = (u + at) dt$

Integrating within the limits $x = 0$ to x and $t = 0$ to t , we get

$$\int_0^x dx = u \int_0^t dt + a \int_0^t t dt$$

or, $[x]_0^x = u [t]_0^t + a [\frac{t^2}{2}]_0^t$

$\therefore x = ut + \frac{1}{2} at^2$

Third equation of motion

For this, we begin with

$$a = v \frac{dv}{dx}$$

$$\Rightarrow a dx = v dv$$

Integrating within the limits $v = u$ to v and $x = 0$ to x , we have

$$a \int_0^x dx = \int_u^v v dv$$

or, $a [x]_0^x = [\frac{v^2}{2}]_u^v$

or, $ax = \frac{1}{2} (v^2 - u^2)$

$\therefore v^2 = u^2 + 2ax$

Distance travelled at t^{th} second

If a body having initial velocity u is uniformly accelerating by a then the distance travelled in t^{th} second is given by

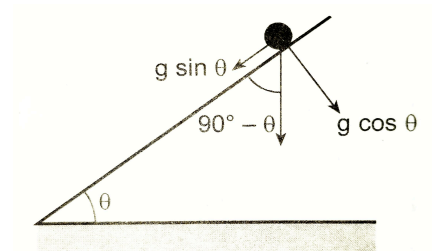
$$s_t = u + \frac{a}{2} (2t - 1)$$

#Motion under gravity

Motion under gravity is vertical motion influenced solely by gravitational force (neglecting air resistance), resulting in constant acceleration of approximately $g = 9.8 \text{ m/s}^2$ downward. When a body falls, the acceleration due to gravity is taken to be positive and if the body is moving up, it is taken to be negative.

Equations for a falling body	Equations for a rising body
$v = u + gt$ $h = ut + \frac{1}{2} gt^2$ $v^2 = u^2 + 2gh$	$v = u - gt$ $h = ut - \frac{1}{2} gt^2$ $v^2 = u^2 - 2gh$

#Motion Down a Smooth Inclined Plane



When an object slides down a smooth (frictionless) inclined plane, it moves under a constant acceleration which is equal to be the acceleration due to gravity times the sine of the angle of the plane with the horizon i.e. $g \sin \theta$.

When a body slides downwards, it is taken to be positive and if the body is moving up, it is taken to be negative.

Equations for body sliding downwards	Equations for body climbing (or moving) upwards
1. $v = u + gt \sin \theta$	1. $v = u - gt \sin \theta$
2. $l = ut + \frac{1}{2}gt^2 \sin \theta$	2. $l = ut - \frac{1}{2}gt^2 \sin \theta$
3. $v^2 = u^2 + 2gl \sin \theta$	3. $v^2 = u^2 - 2gl \sin \theta$